

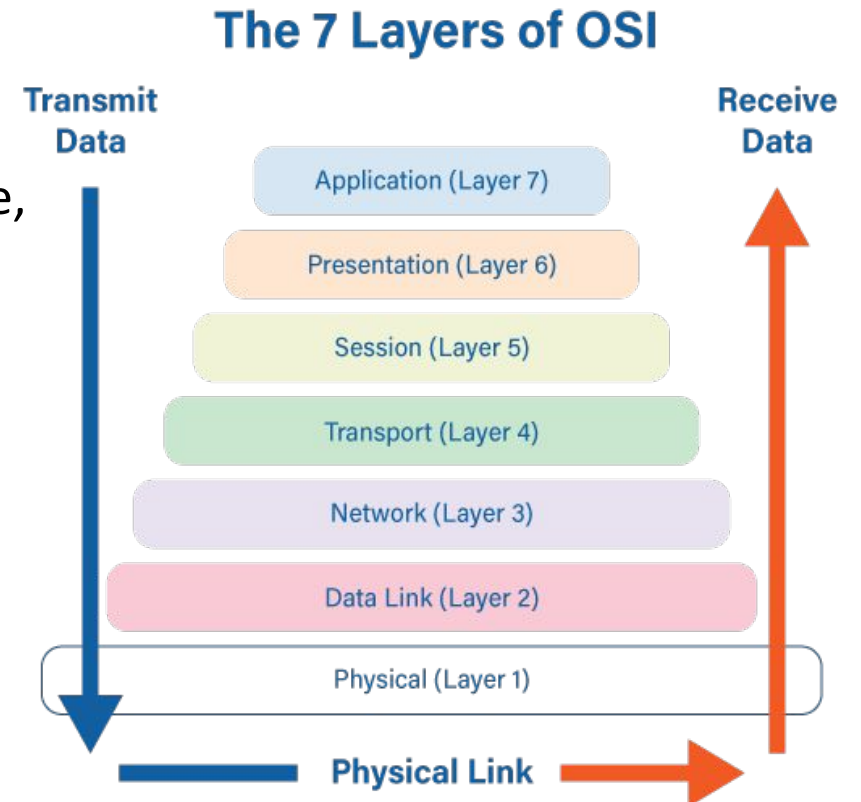
Computer Networks (CS-241)

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Adapted From:

Computer Networking: A Top-Down Approach

8th edition: Jim Kurose, Keith Ross Pearson, 2020

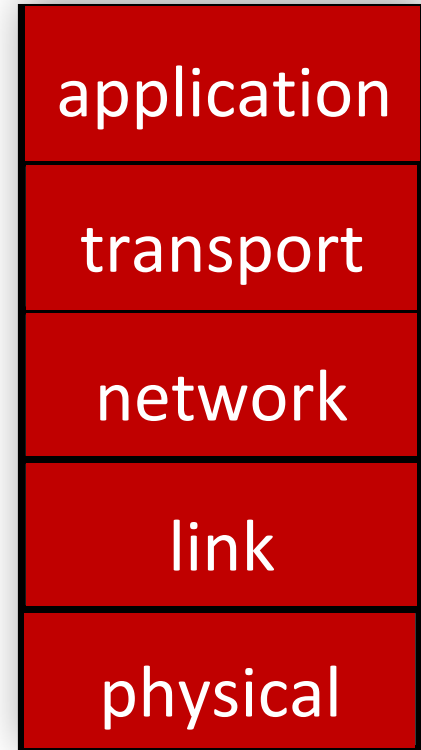


Network layer: our goals

- The Network Layer is responsible for moving data (packets) from the source host to the destination host across multiple networks.
- network layer overview
 - data plane: The Data Plane is the fast, per-packet operation inside routers.
 - control plane: Data plane moves packets according to rules already given.
 - Logical addressing (IP addresses)
 - Routing (choosing paths)
 - Forwarding (moving packets to next hop)
 - Packet handling across different networks

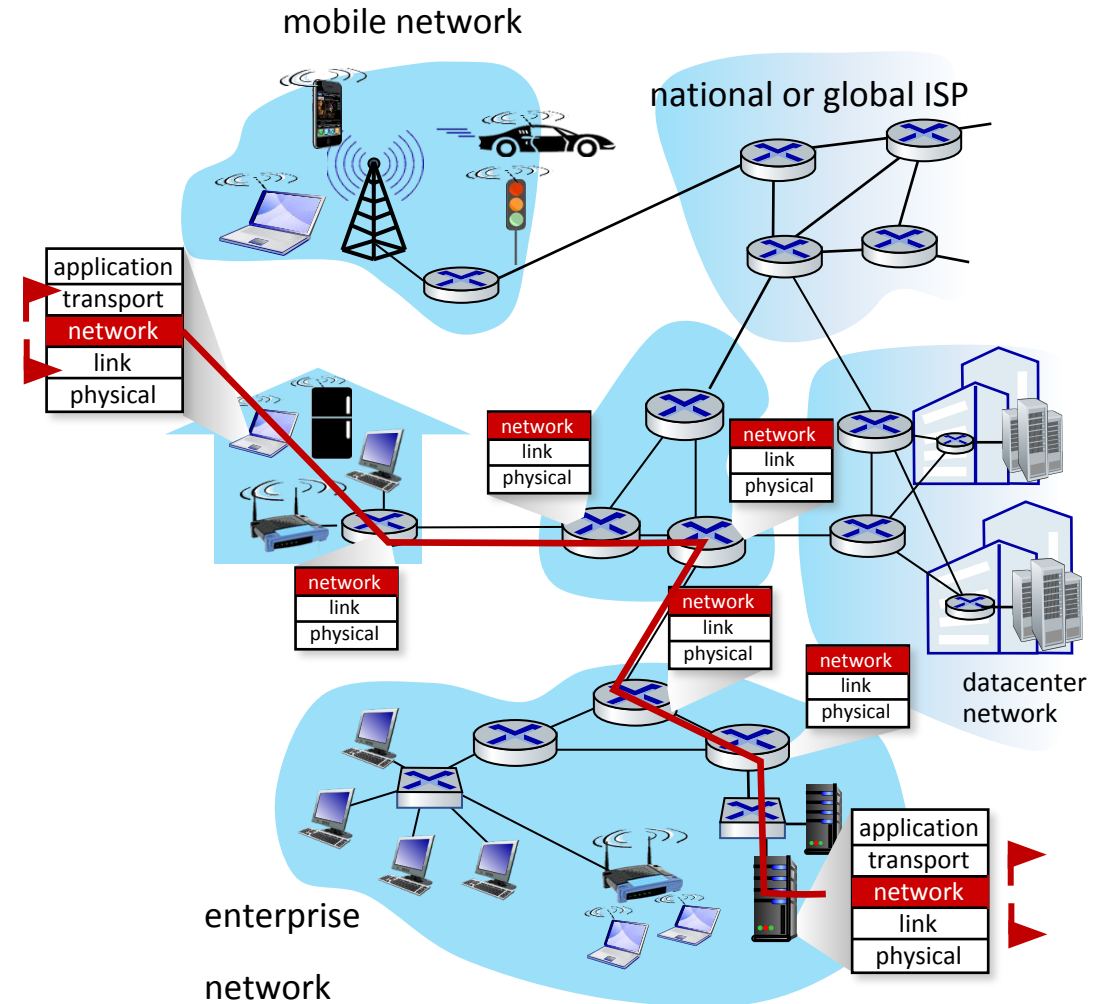
Network layer: “data plane” roadmap

- The Network Layer is the 3rd layer of the TCP/IP model. Its primary responsibility is to provide logical communication between devices located on different networks.
- Unlike the Data Link Layer, which delivers data only within a single local network, the Network Layer enables data to travel across multiple interconnected networks to reach its final destination.
- This layer handles **logical addressing**, **path determination**, and **packet forwarding**. The most widely used Network Layer protocol on the Internet is the **Internet Protocol (IP)**, which assigns unique **IP addresses** to devices and enables global communication.



Network-layer services and protocols

- **transport segment** from sending to receiving host
 - **sender:** encapsulates transport-layer segments into IP datagrams, passes to link layer
 - **receiver:** delivers segments to transport layer protocol
- Network layer protocols **are present in every Internet device**, including **hosts** and **routers**.
- **routers:**
 - examines header fields in all IP datagrams passing through it
 - moves datagrams from input ports to output ports to transfer datagrams along end-end path



Two key network-layer functions

network-layer functions:

- *forwarding*: move packets from a router's input link/port to appropriate router output link/port
- *routing*: determine route taken by packets from source to destination
 - *routing algorithms*
- *local*: single router level decision or actions such as selecting the **output port** for a packet based on the forwarding table.
- *global*: network level decision or actions involving multiple network devices such as **determining the end-to-end path that packets** follow across the network.

analogy: taking a trip

- *forwarding*: process of getting through single interchange
- *routing*: process of planning trip from source to destination (passing through many intersections)



forwarding



routing

Forwarding - explained

- Forwarding is the process of moving a packet from an incoming interface to the appropriate outgoing interface of a router.
 - It is a local and fast operation
 - Performed inside a router
 - Uses a forwarding (routing) table
 - Implemented mostly in hardware for speed
- When a packet arrives at a router, the router examines the destination IP address, looks it up in its forwarding table, and sends the packet to the next hop toward its destination.



Forwarding

Forwarding decides “**where to send the packet next**”, not the full path.

Routing - explained

- Routing is the process of determining the best path that packets should follow from the source to the destination across a network.
 - It is a global and logical process
 - Involves route calculation and path selection
 - Uses routing algorithms and protocols
 - Implemented in software
 - Relatively time consuming
- Routers exchange routing information using routing protocols such as:
 - RIP (Routing Information Protocol)
 - OSPF (Open Shortest Path First)
 - BGP (Border Gateway Protocol)
- Based on this information, routers build and update their routing tables, which are later used for forwarding packets.



Routing

Routing decides “**which path is best**” for packet delivery.

Aspect

Forwarding

Routing

Scope

Local (inside a router)

Global (across networks)

Purpose

Move packet to next hop

Find best path

Speed

Very fast

Relatively slower

Implementation

Hardware-based

Software-based

Decision

Next interface

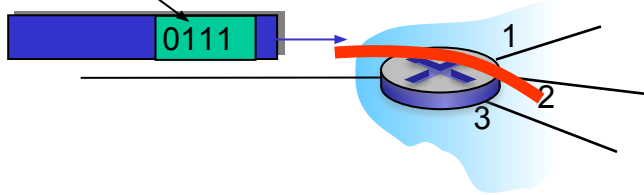
Complete route

Network layer: data plane, control plane

Data plane:

- *local*, per-router function
- determines how datagram arriving on router input port is forwarded to router output port

values in arriving
packet header

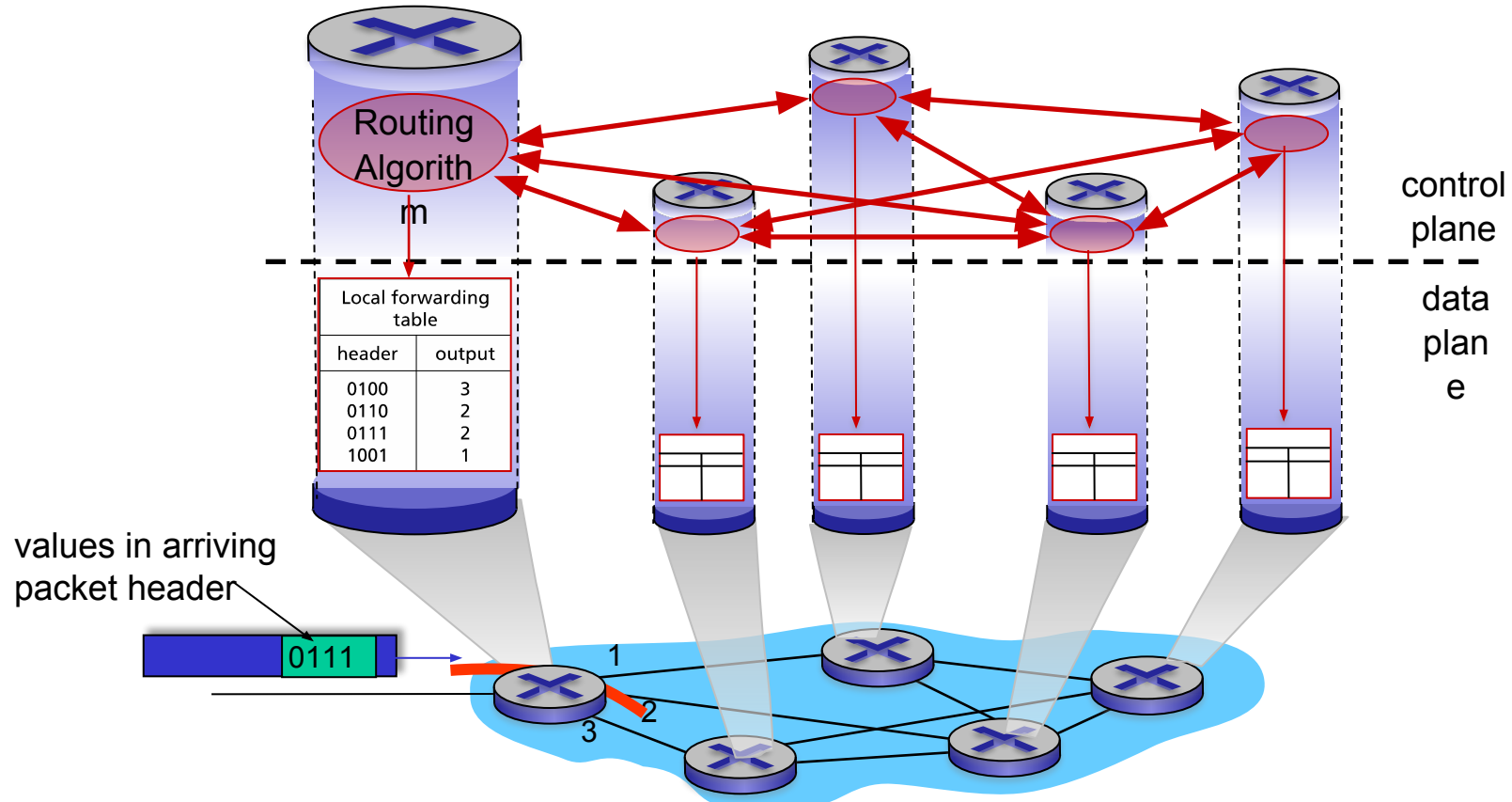


Control plane

- *network-wide* logic
- determines how datagram is routed among routers along end-end path from source host to destination host
- two control-plane approaches:
 - *traditional routing algorithms*: implemented in routers
 - *software-defined networking (SDN)*: implemented in (remote) servers

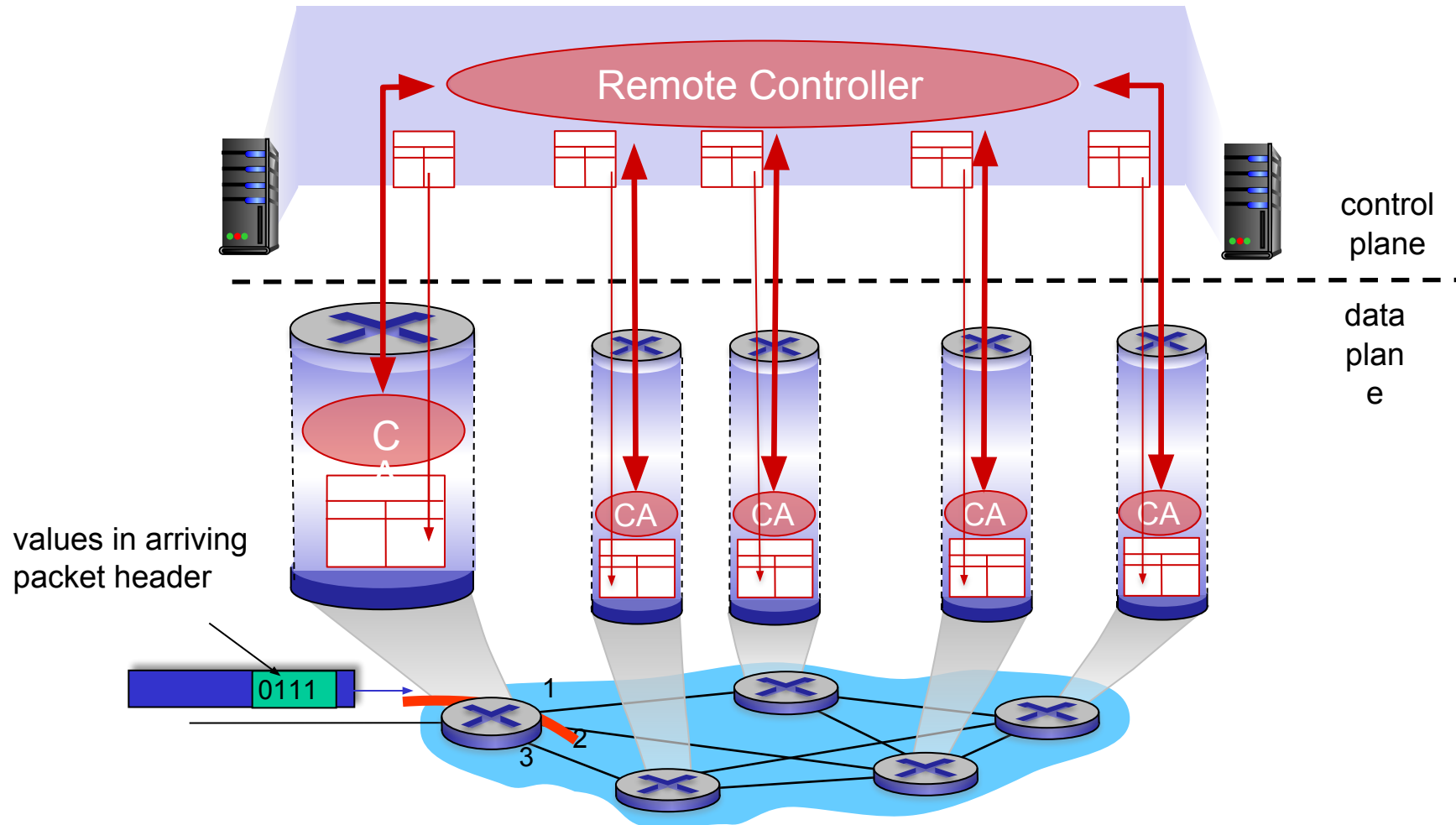
Per-router control plane (traditional routing approach)

- Individual routing algorithm components *in each and every router* interact in the control plane
- Local forwarding table inside each router.
- In traditional routing approach, a distributed routing algorithm runs on every router and all routers communicate with each other to compute values for routing tables



Software-Defined Networking (SDN) control plane

- Remote controller software process computes, installs forwarding tables in each router under its control. Implemented in **data centers, high performance computers** etc.



Network service model

Q: What *service model* for “channel” transporting datagrams from sender to receiver?

A: “Best-effort” = network tries its best, but no strict guarantees. If reliability is needed (like TCP), it’s handled on top of this, at the transport layer.

example services for *individual* datagrams:

- The network ensures that every datagram reaches the destination.
- guaranteed delivery with less than 40 msec delay

example services for a *flow* of datagrams:

- in-order datagram delivery
- guaranteed minimum bandwidth to flow.
i.e. Your video call always gets at least **2 Mbps**, even if the network is busy.
- The time between consecutive packets doesn’t vary too much.

Network-layer service model

Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no

Reflections on best-effort service:

- Even though it's simple, it works well because of:

1. Simplicity:

- Easy to implement, which helped the Internet **grow quickly**.

2. Enough Bandwidth:

- If there is enough network capacity, applications like **video calls or streaming** usually work fine most of the time.

3. **Distributed Services (CDNs, Datacenters):** Services are replicated close to users, so data doesn't have to travel far. **Example:** Netflix uses servers near your city for faster streaming

4. **Congestion Control:** Applications like file transfer adjust their sending rate when the network is congested. This helps the network remain usable and fair for all users.

It's hard to argue with success of best-effort service model

Virtual Circuit Networks (1/2)

A Virtual Circuit Network (VCN) is a network in which a logical path (virtual circuit) is established before data transmission begins. All packets follow the same predefined path from source to destination.

Key Characteristics:

- Connection-oriented
- Path is set up before data transfer
- All packets follow the same route
- Packets arrive in order
- Network maintains state information

Working Phases:

- **Setup Phase** – Virtual circuit is established
- **Data Transfer Phase** – Data packets are sent
- **Teardown Phase** – Virtual circuit is released
- Resources (memory, bandwidth) reserved in routers are freed. Think of it like disconnecting a phone call after the conversation ends.

Virtual Circuit Networks (2/2)

Advantages:

- Predictable performance
- Ordered packet delivery
- Easier congestion and QoS control
 - *Since the path and resources are known, the network can better manage*

Disadvantages:

- Setup delay
 - *Time is needed to establish the virtual circuit before sending data.*
- Failure on path breaks communication
 - *If one router/link fails, the entire virtual circuit is disrupted.*
- Less flexible
 - *Cannot easily adapt to network changes or reroute packets dynamically.*

Examples:

- ATM (Asynchronous Transfer Mode)

Datagram Networks (1/2)

A Datagram Network is a network in which no prior path is established. Each packet (datagram) is treated independently and may take different paths to reach the destination.

Key Characteristics:

- Connection-less
- No setup required
- Packets may follow different routes
- Packets may arrive out of order
- No network state is maintained

Datagram Networks (2/2)

Advantages:

- Fast start (no setup delay)
- Robust and flexible
- Efficient for bursty traffic

Disadvantages:

- No guarantee of order or delivery
- Congestion control is difficult
- Packet loss possible

Examples:

- Internet (IP protocol)
- UDP-based communication

Feature

VCN

Datagram Network

Connection

Connection-oriented

Connection-less

Path setup

Required

Not required

Route

Same for all packets

May vary

Packet order

Guaranteed

Not guaranteed

State info

Maintained by network

Not maintained

Reliability

Higher

Lower

Example

ATM, Frame Relay

Internet (IP)